

RELIANT DIGITECH ELECTRONICS

Data Engineering & Analytics Project

End-to-End Snowflake Medallion Architecture for Retail Intelligence

Developed by Sudhin & Pritam

| Retail Footprint



Leading Multi-City Chain

Reliant DigiTech is a high-growth electronics retailer based in India. We manage high-volume sales across major metros including Bangalore and Hyderabad.

- ✓ 10 Strategic Locations nationwide
- ✓ Premium portfolio: Laptops, TVs, Mobiles
- ✓ Diverse Mall & High-Street store types

Information Source Grid

Source Dataset	Format	Volume	Business Value
Transactional Orders	CSV	100,000+	Revenue & Sales Trends
Customer Feedback	CSV	30,000	Service Quality Metrics
Inventory Records	CSV	1,200	Stock Velocity Analysis
Google Reviews	JSON	15,000	External Sentiment Logic
Store Master Data	Parquet	10	Dimensional Benchmarking

| Technology Ecosystem

Python

Primary driver for data orchestration, file transformation, and connecting to Snowflake using SQLAlchemy and Pandas.

Snowflake

Centralized Data Warehouse managing Bronze, Silver, and Gold layers with high-performance compute clusters.

SQL / SPs

Native Snowflake Stored Procedures utilized for incremental merging, cleansing, and business logic implementation.

Pipeline Architecture

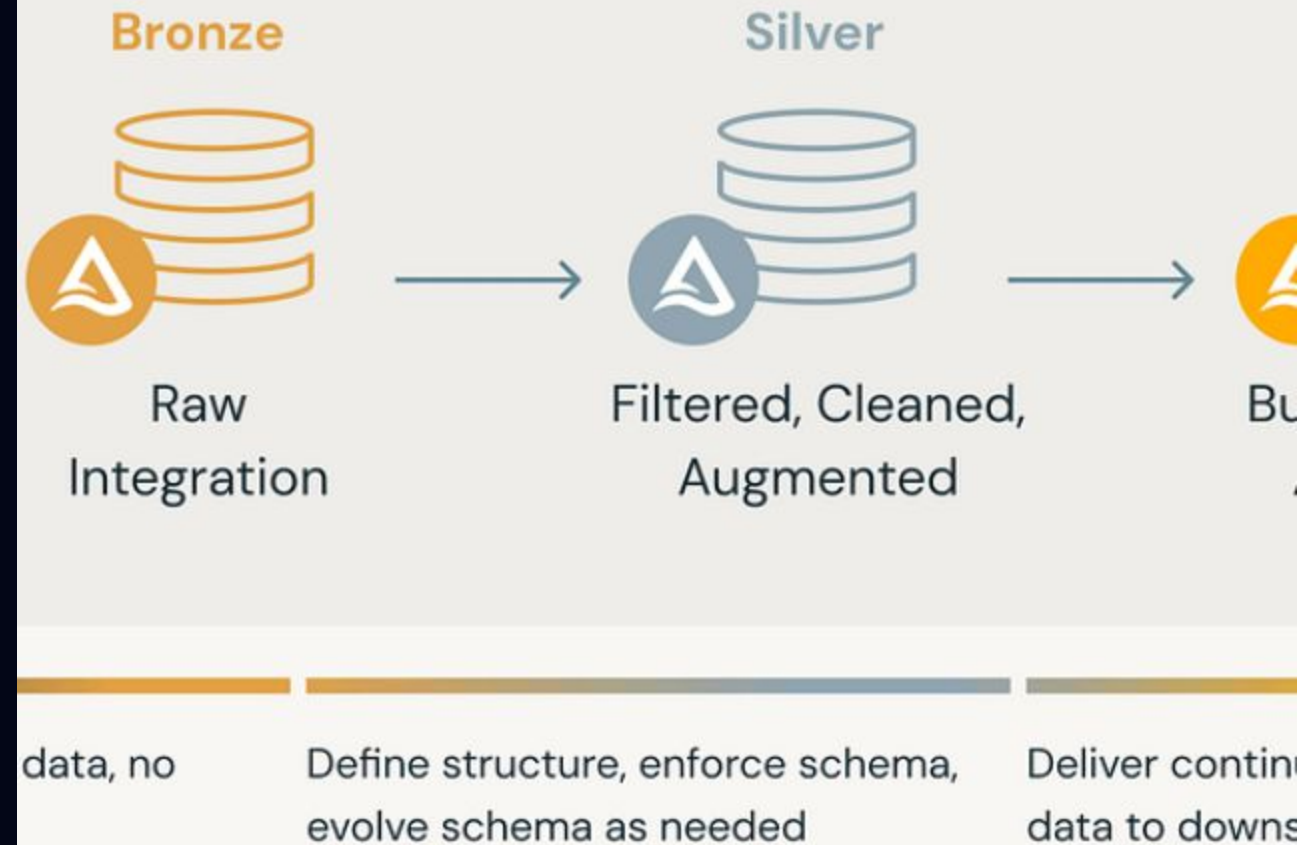
Medallion Design Pattern

We implemented a multi-stage refinery process to convert raw data into high-value business insights.

- ✓ **Bronze Zone:** Raw, immutable landing zone.
- ✓ **Silver Zone:** Validated and merged data.
- ✓ **Gold Zone:** Modeled for reporting (Star Schema).

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| Data Modeling

Fact Tables

Quantitative transactional data including Sales, Inventory snapshots, and Customer Sentiment scores.

Dimension Tables

Descriptive attributes for Store details, Product categories, and Customer demographics used for filtering.

Implementing a Star Schema for optimized Query performance in BI tools.

| Generated Insights

Sales & Revenue

Identifying top 10 products, city-wise distribution, and holiday impact on electronics demand.

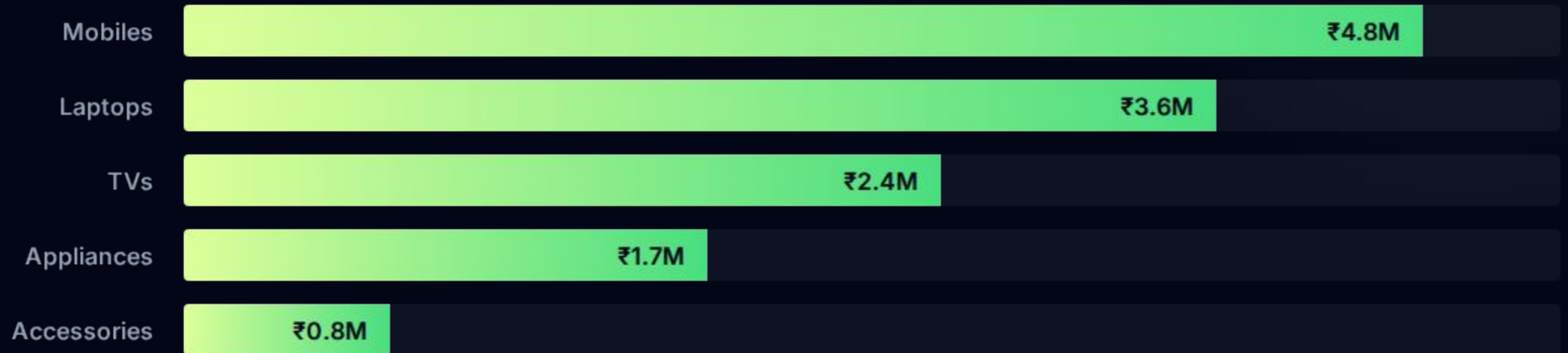
Inventory Health

Monitoring closing stock levels and identifying low-velocity SKUs to optimize capital allocation.

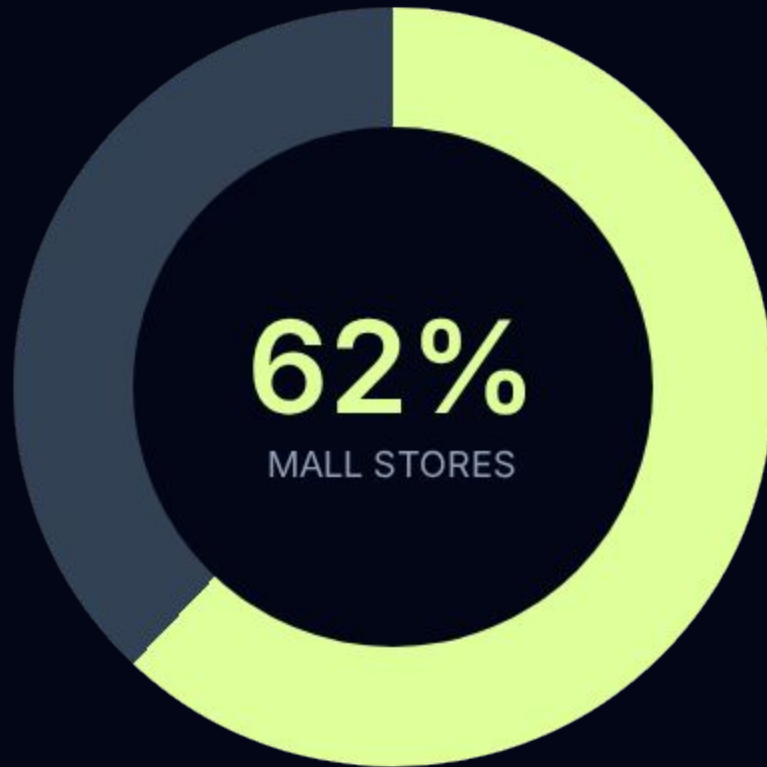
Customer DNA

Analyzing ratings distribution (1-5 stars) and benchmarking external vs. internal feedback scores.

| Product Performance



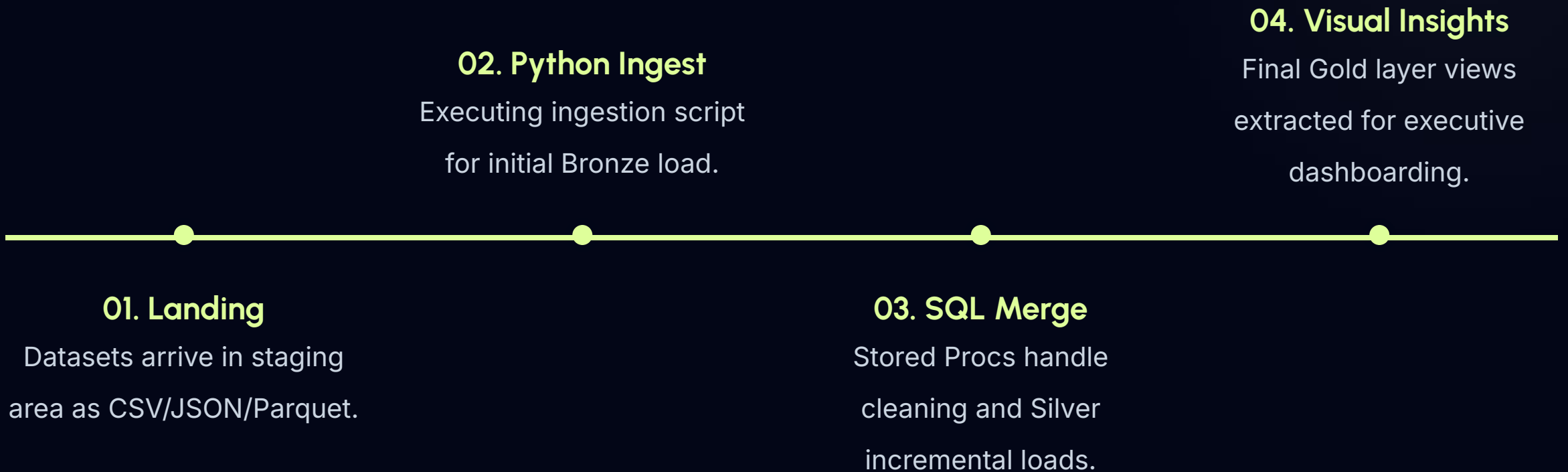
| Store Type Analysis



■ Mall Stores (62%)

■ High Street (38%)

| Pipeline Orchestration



| Core Competencies

Data Engineering

- ✓ Medallion Architecture Design
- ✓ Incremental Watermarking
- ✓ Star Schema Implementation

Data Analysis

- ✓ 100k+ Transaction Analysis
- ✓ Customer Segmentation
- ✓ Sentiment Benchmarking

Project Finalized

Successfully engineered a robust, scalable retail analytics platform using modern cloud data tools.

Thank You |

Questions?

Sudhin & Pritam | Reliant DigiTech Electronics

| Image Sources



https://images.stockcake.com/public/8/4/3/84371ed1-2aa8-414e-8b70-7d2c39d8af7f_large/modern-electronics-store-stockcake.jpg

Source: stockcake.com



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